Quantitative Research Methods in HCI

Human-Computer Interaction

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Why Quantitative Research?

Sometimes you need numbers to make decisions

Measuring performance and patterns

Comparing alternatives objectively

Generalizing findings to larger populations

Today's Journey

1. **Understanding Quantitative Research** — Philosophy, strengths, and when to use it
2. **Survey Research** — Questionnaire design, sampling, and administration
3. **Experimental Design** — Variables, conditions, and controlled comparisons
4. **Statistical Analysis** — Hypothesis testing, significance, and effect sizes
5. **Quality & Ethics** — Validity, power, and ethical considerations

Goal

Learn to design rigorous quantitative studies that measure, compare, and generalize

What is Quantitative Research?

The Quantitative Approach

Quantitative research involves collecting and analyzing numerical data to:

- Measure variables
- Test hypotheses
- Identify patterns
- Make predictions
- Generalize findings

Core characteristics:

- Structured, predetermined approach
- Standardized data collection
- Statistical analysis
- Objective measurements
- Larger sample sizes
- Replicable procedures

> From qualitative depth (understanding) to quantitative breadth (measurement)

When to Use Quantitative Research

Quantitative research is ideal for:

- Testing specific hypotheses
- Measuring performance (speed, accuracy, errors)
- Comparing alternatives (which is better?)
- Identifying statistical patterns
- Generalizing to populations
- Establishing causality (experiments)
- Tracking changes over time

Examples in HCI:

- “Is touch typing faster than hunt-and-peck?”
- “Does dark mode reduce eye strain?”
- “What percentage of users complete onboarding?”
- “Which menu organization leads to faster task completion?”

> Use quantitative methods when you need to measure, compare, or test at scale

Qualitative to Quantitative

Yesterday: Qualitative

- Explores meaning, experiences
- Open-ended, flexible
- Rich, detailed descriptions
- Small samples, depth
- **Why? How? What does it mean?**
- Words, images, observations

Today: Quantitative

- Measures variables, tests hypotheses
- Structured, predetermined
- Statistical patterns
- Larger samples, breadth
- **How many? How much? How often?**
- Numbers, measurements

> Both are valuable! Often used together for comprehensive understanding

Two Main Quantitative Methods in HCI

Surveys

- Collect self-reported data
- Attitudes, preferences, behaviors
- Large samples possible
- Structured questionnaires
- Statistical patterns

Strengths:

- Cost-effective
- Can reach many people
- Standardized data

Limitations:

- Self-report bias
- Response rates
- Can’t establish causality

Experiments

- Controlled comparisons
- Manipulate variables
- Measure outcomes
- Establish causality
- Precise measurements

Strengths:

- Control over variables
- Can determine causality
- Fine-grained data

Limitations:

- Artificial settings
- Time-intensive
- Smaller samples

Survey Research

What Are Surveys Good For?

Surveys help you understand:

- Attitudes and opinions
- Self-reported behaviors
- Demographics and characteristics
- Satisfaction and preferences
- Awareness and knowledge

Types of surveys:

- **Descriptive** — What's the distribution? How common?
- **Analytic** — What relationships exist? What predicts what?

Not well-suited for:

- Objective performance measures
- Establishing causality
- In-depth understanding

> Surveys provide structured self-reported data from potentially large samples

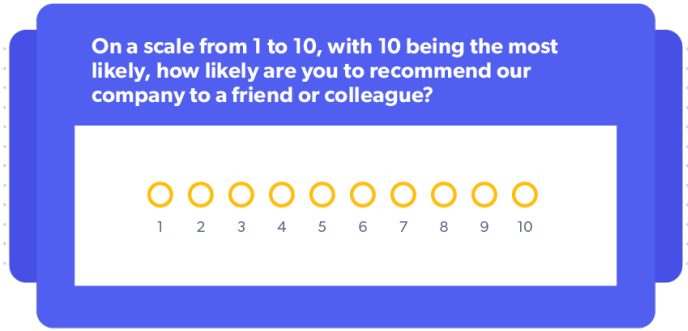
Planning a Survey Study

1. **Define research focus**
   - What do you want to learn?
   - What constructs will you measure? (satisfaction, usability, trust, etc.)
2. **Choose questionnaire**
   - Use validated questionnaire when possible (more on this next lecture!)
   - Or design new one (requires careful testing)
3. **Plan sampling**
   - Who to survey?
   - How many? (see power analysis)
   - How to reach them?
4. **Pilot test**
   - Test with small group first
   - Check clarity, timing, comprehension
5. **Collect and analyze**
   - Administer survey
   - Clean and analyze data
   - Draw conclusions carefully

Designing Questions: The Basics

Good question characteristics:

- Clear and unambiguous
- One idea per question
- Appropriate vocabulary
- Neutral (not leading)
- Answerable by respondent



Question types:

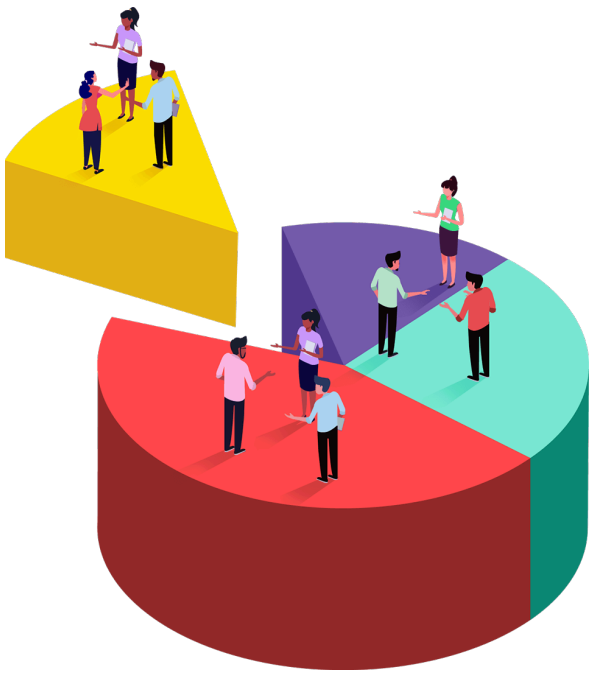
- **Closed-ended** — Fixed response options (Yes/No, Multiple choice, Rating scales)
- **Open-ended** — Free-text response (use sparingly, harder to analyze)

Likert scales (most common in HCI):

- Usually 5 or 7 points
- Example: “The system was easy to use”
  - 1=Strongly Disagree, 2=Disagree, 3=Neutral, 4=Agree, 5=Strongly Agree

> Avoid: leading questions, double-barreled questions, jargon, vague terms

Sampling for Surveys



Sampling terminology:

- **Population** — All potential users
- **Sampling frame** — Those you can contact
- **Sample** — Those you invite
- **Respondents** — Those who actually complete survey

Sampling strategies:

- **Random** — Every member has equal chance
- **Stratified** — Ensure representation of subgroups
- **Convenience** — Use available people (common but problematic)
- **Snowball** — Participants recruit others

> Goal: Representative sample that lets you generalize findings

Sample size: HCI surveys average ~371 respondents, but varies widely (6 to 50,000!)

Response Rates and Bias

Response rates are often low:

- Online surveys: 10-30% typical
- Email surveys: 20-40%
- In-person: higher

Non-response bias: Those who respond may differ from those who don't

Strategies to improve response:

- Keep survey short (< 10 minutes)
- Clear, compelling invitation
- Explain why it matters
- Offer incentive (but not coercive)
- Send reminders
- Make it mobile-friendly
- Show progress indicator

> Even with high response, consider who chose to participate vs. who didn't

Examples from Research

Large-Scale Survey Studies

Herbert, F., Becker, S., Schaewitz, L., Hielscher, J., Kowalewski, M., Sasse, A., Acar, Y., & Dürmuth, M. (2023). A World Full of Privacy and Security (Mis)conceptions? Findings of a Representative Survey in 12 Countries. **CHI '23**, Article 582.

Surveyed 12,351 participants across 12 countries using quota-representative sampling aligned with census data. Investigated security and privacy misconceptions with rigorous pilot testing, pre-registration, and statistical analysis using regression models. Demonstrates cross-cultural survey design and how to handle complexity of multi-country comparisons.

Validated Questionnaire Development

Davis, F. D. (1989). Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology. **MIS Quarterly**, 13(3), 319-340.

Developed the Technology Acceptance Model (TAM) scales through systematic validation: literature review of 37 papers, generated 14 candidate items per construct, refined through expert review and pilot testing. Two empirical studies validated the scales achieving Cronbach's alpha of .94-.98. Over 60,000 citations demonstrate how rigorous scale development leads to lasting impact.

Experimental Design

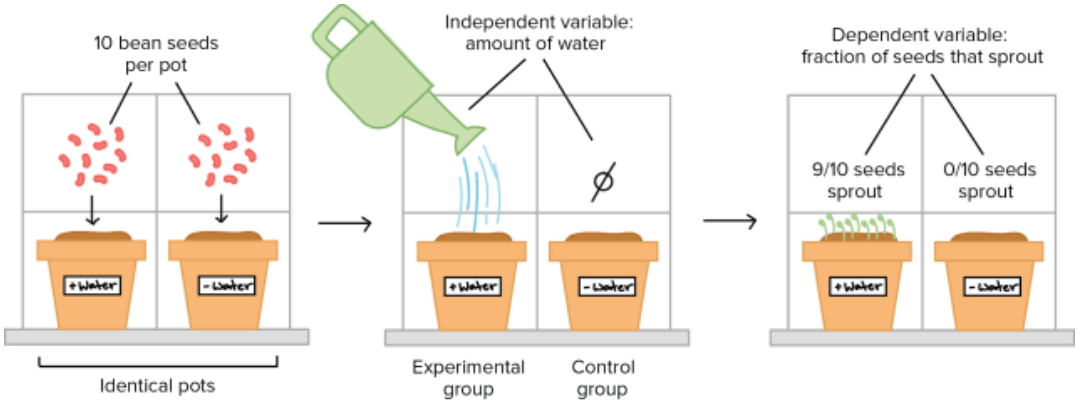
Why Experiments?

Experiments let you:

- Compare alternatives systematically
- Control extraneous factors
- Establish cause-and-effect relationships
- Make precise measurements
- Understand mechanisms of interaction

When to use experiments:

- Testing specific hypotheses
- Comparing interface designs
- Measuring performance (speed, accuracy)
- Understanding interaction techniques
- Evaluating algorithms



> Experiments provide controlled comparisons and can establish causality

Variables: The Building Blocks

Independent variables (IV)

- What you manipulate
- The “cause”
- Under your control

Examples:

- Input method (touch vs. mouse)
- Interface layout (grid vs. list)
- Font size (12pt vs. 16pt)

Levels: Different values of IV

- Mouse vs. Touchpad vs. Trackpoint

Dependent variables (DV)

- What you measure
- The “effect”
- Outcomes of interest

Examples:

- Task completion time
- Error rate
- Satisfaction rating
- Clicks to complete task

Should be: Objective, reliable, relevant

> Hypothesis: IV affects DV — “Touch input will be faster than mouse input”

Experimental Design: Between vs. Within

Between-subjects

- Each participant experiences ONE condition
- Different groups for each condition

Advantages:

- No learning effects
- No fatigue
- Simpler analysis

Disadvantages:

- Need more participants
- Individual differences confound
- Less statistical power

Example: Group A uses Interface 1, Group B uses Interface 2

Within-subjects (repeated measures)

- Each participant experiences ALL conditions
- Same people, multiple conditions

Advantages:

- Fewer participants needed
- Control for individual differences
- More statistical power

Disadvantages:

- Learning effects
- Fatigue
- Need to counterbalance order

Example: Everyone uses both Interface 1 and Interface 2 (order varied)

> Within-subjects is more common in HCI due to individual variability in performance

Confounds and Controls

Confounding variable: Something that affects DV but isn't the IV

Examples:

- Prior experience with similar interfaces
- Time of day (morning vs. evening)
- Noise in environment
- Individual differences (age, expertise)

How to control confounds:

- **Randomization** — Randomly assign participants to conditions
- **Counterbalancing** — Vary order of conditions
- **Standardization** — Keep everything else constant
- **Blocking** — Group by confounding variable
- **Statistical control** — Measure and account for in analysis

> Good experiments minimize confounds so you can be confident about cause-effect

Hypotheses: Making Predictions

Hypothesis: Testable prediction about relationship between variables

Two types:

- **Null hypothesis (H_0)** — No effect, no difference
- **Alternative hypothesis (H_1)** — There is an effect

Example:

- **H_0**: There is no difference in task completion time between touch and mouse input
- **H_1**: Task completion time differs between touch and mouse input

Or directional:

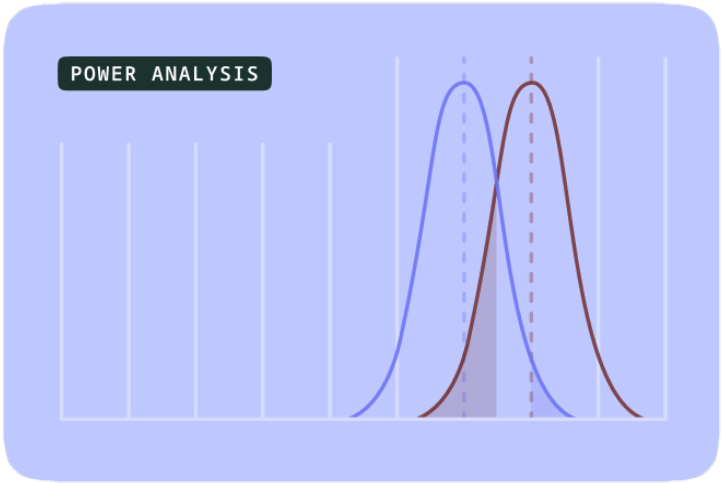
- **H_1**: Touch input will be faster than mouse input

> Statistics help you decide whether to reject H_0 in favor of H_1

Sample Size and Power

Power analysis helps determine sample size based on:

- Expected effect size (how big is the difference?)
- Desired statistical power (typically 80%)
- Significance level (typically $\alpha = 0.05$)
- Study design (between vs. within)



Typical HCI studies:

- 12 participants average (all studies)
- 20 for controlled experiments
- More for between-subjects designs

> For medium effect size, 80% power: need 64 participants PER GROUP for between-subjects!

Reality: HCI studies often underpowered — be aware of this limitation

Examples from Research

Controlled Experiment: Input Device Comparison

MacKenzie, I. S., Sellen, A., & Buxton, W. (1991). A comparison of input devices in elemental pointing and dragging tasks. **CHI '91**, 161-166.

Classic within-subjects design with 12 participants comparing mouse, tablet/stylus, and trackball performance across pointing and dragging tasks. Demonstrates exemplary experimental control: device order fully counterbalanced across participants, task order randomized and alternated across sessions, 16 amplitude-width conditions randomized within each session. Practice sessions preceded each condition to control learning effects. Measured both speed (movement time) and accuracy (error rate), revealing that mouse and tablet achieved 670ms pointing times versus 1100ms for trackball. The study established methodology later codified in ISO 9241-9 for pointing device evaluation—showing how rigorous experimental design leads to lasting standards.

Statistical Analysis

Why Statistics?

Statistics help us:

- Describe data (means, distributions)
- Account for variability and uncertainty
- Determine if differences are “real” or just chance
- Make inferences about populations from samples
- Quantify relationships between variables

Two main types:

- **Descriptive statistics** — Summarize data (mean, median, SD, graphs)
- **Inferential statistics** — Draw conclusions about populations

> Every observation has variability — statistics help us understand what we can conclude despite uncertainty

Descriptive Statistics: Start Here

Measures of central tendency:

- **Mean** — Average (sensitive to outliers)
- **Median** — Middle value (robust)
- **Mode** — Most common value

Measures of spread:

- **Standard deviation** — Average distance from mean
- **Range** — Min to max
- **Interquartile range** — Middle 50%

Visualizations:

- **Histograms** — Distribution of values
- **Box plots** — Median, quartiles, outliers
- **Scatter plots** — Relationships between variables
- **Bar charts** — Means with error bars

> Look at your data! Check for outliers, skew, unusual patterns

Before inferential statistics: Understand distribution, check assumptions

Hypothesis Testing: The Logic

1. Assume null hypothesis is true (no difference)
2. Calculate: How likely are results this extreme **if H_0 is true**?
3. If very unlikely ($p < 0.05$), reject H_0
4. If not unlikely enough, fail to reject H_0

p-value

Probability of getting results this extreme (or more) if null hypothesis is true

Significance level (α):

- Common threshold: $p < 0.05$
- Means: < 5% chance result is due to randomness
- Also see: $p < 0.01$, $p < 0.001$ for stronger evidence

$p < 0.05$ doesn't mean "proven" — it means "unlikely to be just chance"

Common Statistical Tests

t-test Compare means of two groups

- **Independent samples t-test** — Between-subjects
- **Paired samples t-test** — Within-subjects

Example Is task time faster with Interface A vs. B?

Assumptions Normal distribution, similar variances

For three or more groups ANOVA (Analysis of Variance)

When assumptions not met Non-parametric alternatives exist (Mann-Whitney U, Wilcoxon signed-rank)

> Choose the appropriate test based on your design and data characteristics

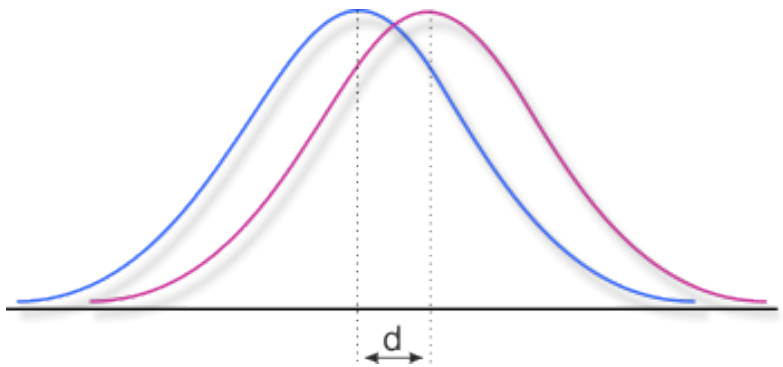
Effect Size: How Big Is the Difference?

p-value tells you: Is there likely a difference?

Effect size tells you: How big is the difference? (More important!)

Common effect size measures:

- **Cohen's d** — Standardized mean difference
  - Small: 0.2, Medium: 0.5, Large: 0.8
- **Partial eta squared (ηp^2)** — For ANOVA
- **r** — For correlations



Why effect size matters:

- Statistical significance ≠ practical significance
- Large sample can make tiny differences "significant"
- Small sample might miss large effects

> Always report both p-values AND effect sizes!

Quality & Ethics

Validity in Experiments

1. **Internal validity** — Does IV actually cause DV?
   - Threatened by confounds
   - Control through randomization, standardization
   - Most important for experiments
2. **External validity** — Do results generalize?
   - To other people, settings, times
   - Often trade-off with internal validity (lab control vs. realism)

> Strong experiments maximize both — but trade-offs exist!

Common Threats to Validity

Selection bias Non-representative sample

Demand characteristics Participants guess purpose, behave accordingly

Experimenter effects Researcher unconsciously influences results

Hawthorne effect Behavior changes because being observed

Practice effects Learning over repeated trials

Fatigue effects Performance declines over time

Regression to mean Extreme scores become less extreme

> Design experiments to minimize these threats

Ethical Considerations in Experiments

1. **Informed consent**
   - Explain purpose, procedures, risks
   - Voluntary participation
   - Right to withdraw anytime
2. **No harm**
   - Physical, mental, or emotional
   - Minimize discomfort
   - Monitor participants
3. **Privacy and confidentiality**
   - Anonymize data
   - Secure storage
   - Only share what's necessary
4. **Fair compensation**
   - Adequate for time
   - Not coercive

> Ethics is not just IRB approval — it's an ongoing responsibility

Reporting Quantitative Research

1. **Introduction** — Research question, hypotheses, rationale
2. **Related work** — Previous findings, theoretical background
3. **Method**
   - Participants (n, demographics, recruitment)
   - Materials (interfaces, questionnaires)
   - Procedure (step-by-step, including instructions)
   - Design (IVs, DVs, between/within)
4. **Results**
   - Descriptive statistics
   - Inferential tests
   - Effect sizes
   - Visualizations
5. **Discussion** — Interpret findings, limitations, implications

> Transparency: Report everything needed for replication!

Common Pitfalls in Quantitative Research

1. **p-hacking** — Trying many analyses until finding $p < 0.05$
2. **HARKing** — Hypothesizing After Results are Known
3. **Cherry-picking** — Only reporting significant results
4. **Ignoring assumptions** — Using tests when assumptions violated
5. **Small samples** — Underpowered studies
6. **Confusing correlation and causation** — Regression ≠ causation
7. **Over-interpreting p-values** — $p < 0.05$ doesn't mean "proven"
8. **Multiple comparisons** — Inflated Type I error rate
9. **Missing data** — Not handling appropriately

> Pre-register studies when possible to prevent these issues!

Practical Tips for Success

1. **Always pilot test** — Find problems before full study
2. **Plan analysis before collecting data** — Avoid p-hacking
3. **Use validated measures** — Better reliability
4. **Document everything** — Enable replication
5. **Visualize your data** — Understand patterns
6. **Report effect sizes** — Not just p-values
7. **Be honest about limitations** — Every study has them
8. **Seek statistical consultation** — When in doubt
9. **Share data and materials** — Transparency and reproducibility

> Rigorous quantitative research requires careful planning and execution

Quantitative + Qualitative

Mixed methods provide:

- Quantitative breadth + qualitative depth
- “What” and “how much” + “why” and “how”
- Measurement + understanding

Example: Evaluating a new interface

- **Quantitative**: Measure task time, error rate, satisfaction
- **Qualitative**: Interview users about experience, frustrations
- **Result**: Know it’s slower (numbers) and why (confusing navigation)

> Don’t limit yourself to one paradigm — use the right tool for each question!

Key Takeaways

1. **Quantitative research provides measurement and generalization** — Tests hypotheses with statistical rigor
2. **Two main methods: surveys and experiments** — Surveys for attitudes/behaviors; experiments for causality
3. **Validated questionnaires exist** — Use them when possible (more on this next lecture!)
4. **Experiments require careful design** — Control confounds, adequate power, appropriate measures
5. **Variables are fundamental** — Independent (manipulated) and dependent (measured)
6. **Statistics manage uncertainty** — Descriptive then inferential; report p-values AND effect sizes
7. **Validity matters** — Internal (causality) and external (generalization)
8. **Sample size matters** — Do power analysis; typical HCI studies may be underpowered
9. **Ethics is paramount** — Informed consent, no harm, privacy, fair compensation
10. **Transparency enables scrutiny** — Document and share methods, data, analysis

> Next lecture: How to properly measure psychological constructs (psychometrics)

Resources

Essential reading:

- Lazar, J., Feng, J. H., & Hochheiser, H. (2017). **Research Methods in Human-Computer Interaction** (2nd ed.). Morgan Kaufmann.
- Field, A. (2017). **Discovering Statistics Using IBM SPSS Statistics** (5th ed.). SAGE.
- Robertson, J., & Kaptein, M. (2016). **Modern Statistical Methods for HCI**. Springer.

Survey design:

- Sauro, J., & Lewis, J. R. (2016). **Quantifying the User Experience** (2nd ed.). Morgan Kaufmann.
- Fink, A. (2016). **How to Conduct Surveys** (6th ed.). SAGE.

Experimental design:

- Cairns, P., & Cox, A. L. (Eds.). (2008). **Research Methods for Human-Computer Interaction**. Cambridge University Press.

Tools:

- R, SPSS, Jamovi (statistical analysis)
- G*Power (power analysis)
- Qualtrics, Google Forms (survey platforms)

Questions?